Avance Proyecto Semana 11

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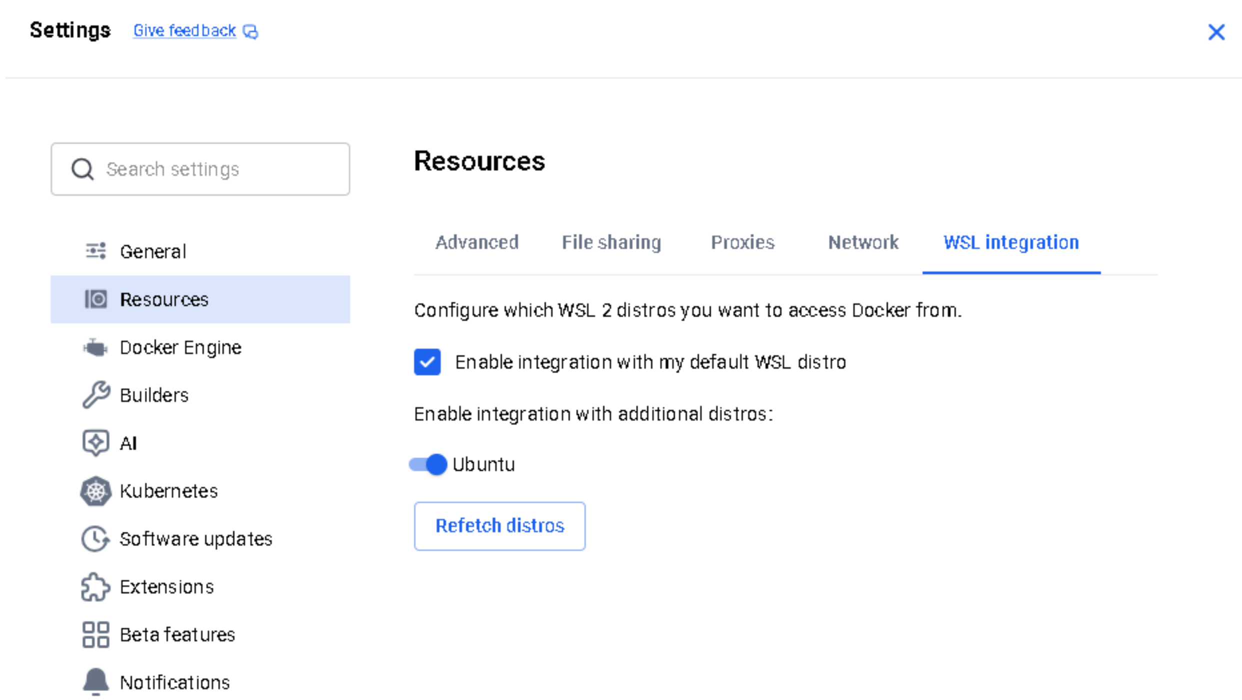
URLs principales

- Nginx: http://localhost
- Prometheus: http://localhost:9090
- Loki: http://localhost:3100
- N8N: http://localhost:5678

Paso 0: Integrar Docker Desktop con WSL2



PASO 1: Instalar Docker dentro de WSL (NO Docker Desktop):

1. Actualizar paquetes:

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo apt update && sudo apt upgrade -y
```

2. Instalar dependencias necesarias

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo apt install -y apt-transport-https ca-certificates curl software-properties-common
age lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203~24.04.1).
```

3. Actualizar e instalar Docker

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo apt update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

4. Verificar instalación

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo apt install -y docker.io docker-compose
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
```

5. Versiones del Docker y Docker compose

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ docker --version
Docker version 29.6.2, build dfc4efb
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ docker-compose --version
Docker Compose version v5.3.1
```

Paso 2 — Verificar que WSL2 use systemd

Esto es importante porque fail2ban y algunos servicios funcionan mejor con systemd.

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ ps -p 1 -o comm=
systemd
```

PASO 3: Activar servicios e instalar herramientas de hardening

1. Activar Docker

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo systemctl enable --now docker
[sudo] password for julio:
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

2. Verificar que Docker está corriendo

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-07-20 20:01:19 CST; 1h 14min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 15533 (dockerd)
      Tasks: 10
     Memory: 25.4M (peak: 74.0M)
        CPU: 4.722s
     CGroup: /system.slice/docker.service
             └─15533 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jul 20 20:01:17 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:17.918622016-06:00" level=info msg="Restoring containers: start."
Jul 20 20:01:18 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:18.019919114-06:00" level=info msg="Deleting nftables IPv4 rules… status 1"
Jul 20 20:01:18 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:18.054790364-06:00" level=info msg="Deleting nftables IPv6 rules… status 1"
Jul 20 20:01:18 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:18.747603469-06:00" level=info msg="Loading containers: done."
Jul 20 20:01:18 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:18.769721400-06:00" level=info msg="Docker daemon" commit=3d8046…ion=29.6.2
Jul 20 20:01:18 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:18.770327571-06:00" level=info msg="Initializing buildkit"
Jul 20 20:01:19 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:19.149173813-06:00" level=info msg="Completed buildkit initialization"
Jul 20 20:01:19 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:19.158624328-06:00" level=info msg="Daemon has completed initialization"
Jul 20 20:01:19 DESKTOP-JN2R04E dockerd[15533]: time="2026-07-20T20:01:19.158759044-06:00" level=info msg="API listen on /run/docker.sock"
Jul 20 20:01:19 DESKTOP-JN2R04E systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
```

3. Instalar herramientas necesarias de hardening

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo apt install -y git ufw fail2ban curl nano
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.11).
nano is already the newest version (7.2-2ubuntu0.2).
nano set to manually installed.
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
```

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ fail2ban-client --version
Fail2Ban v1.0.2
```

4. Verificar fail2ban

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo systemctl status fail2ban --no-pager
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-07-20 21:23:59 CST; 8min ago
       Docs: man:fail2ban(1)
   Main PID: 16778 (fail2ban-server)
      Tasks: 5 (limit: 4612)
     Memory: 22.1M (peak: 65.1M)
        CPU: 655ms
     CGroup: /system.slice/fail2ban.service
             └─16778 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Jul 20 21:23:59 DESKTOP-JN2R04E systemd[1]: Started fail2ban.service - Fail2Ban Service.
Jul 20 21:23:59 DESKTOP-JN2R04E fail2ban-server[16778]: 2026-07-20 21:23:59,791 fail2ban.configreader   [16778]: WARNING 'allowipv6' not…e: 'auto'
Jul 20 21:24:00 DESKTOP-JN2R04E fail2ban-server[16778]: Server ready
Hint: Some lines were ellipsized, use -l to show in full.
```

Paso 4: Configurar (UFW y fail2ban)

Ahora configuramos el firewall. Para el avance, abrimos solo los puertos necesarios del proyecto.

1. el firewall ahora bloquea todo el tráfico entrante por defecto

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
```

- Ese mensaje es normal y esperado. Significa que el firewall ahora bloquea todo el tráfico entrante por defecto (lo cual es seguro), y te está recordando que debes permitir explícitamente los puertos que quieras abrir (que es justo lo que vamos a hacer ahora).

2. el firewall ahora permite todo el tráfico que sale del servidor

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
```

3. Estos puertos corresponden a los servicios base

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 80/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 3306/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 5678/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 9090/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 9100/tcp
Rules updated
Rules updated (v6)
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw allow 3100/tcp
Rules updated
Rules updated (v6)
```

4. Activar el firewall

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw --force enable
Firewall is active and enabled on system startup
```

5. Muestra el estado actual del firewall

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
3306/tcp                   ALLOW IN    Anywhere
5678/tcp                   ALLOW IN    Anywhere
9090/tcp                   ALLOW IN    Anywhere
9100/tcp                   ALLOW IN    Anywhere
3100/tcp                   ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
3306/tcp (v6)              ALLOW IN    Anywhere (v6)
5678/tcp (v6)              ALLOW IN    Anywhere (v6)
9090/tcp (v6)              ALLOW IN    Anywhere (v6)
9100/tcp (v6)              ALLOW IN    Anywhere (v6)
3100/tcp (v6)              ALLOW IN    Anywhere (v6)
```

Paso 5: Crear carpeta del proyecto

1. Creamos la carpeta de nutrialianza

```
julio@DESKTOP-JN2R04E:/mnt/c/Windows/system32$ cd ~
julio@DESKTOP-JN2R04E:~$ mkdir -p nutrialianza-monitoreo
julio@DESKTOP-JN2R04E:~$ cd nutrialianza-monitoreo/
```

2. Ahora creamos la estructura:

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ mkdir -p nginx/html mysql/init prometheus loki filebeat n8n logs/nginx
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch docker-compose.yml .env .env.example README.md
```

3. Archivos para cada carpeta para luego usarlos con nano

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch nginx/html/index.html nginx/default.conf
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch mysql/init/01_schema.sql
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch prometheus/prometheus.yml
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch loki/loki-config.yml
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ touch filebeat/filebeat.yml
```

4. Mostramos las carpetas

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ ls -la
total 36
drwxr-xr-x 9 julio julio 4096 Jul 20 22:07 .
drwxr-x--- 5 julio julio 4096 Jul 20 22:06 ..
-rw-r--r-- 1 julio julio    0 Jul 20 22:07 .env
-rw-r--r-- 1 julio julio    0 Jul 20 22:07 .env.example
-rw-r--r-- 1 julio julio    0 Jul 20 22:07 README.md
-rw-r--r-- 1 julio julio    0 Jul 20 22:07 docker-compose.yml
drwxr-xr-x 2 julio julio 4096 Jul 20 22:18 filebeat
drwxr-xr-x 3 julio julio 4096 Jul 20 22:07 logs
drwxr-xr-x 2 julio julio 4096 Jul 20 22:18 loki
drwxr-xr-x 3 julio julio 4096 Jul 20 22:07 mysql
drwxr-xr-x 2 julio julio 4096 Jul 20 22:07 n8n
drwxr-xr-x 3 julio julio 4096 Jul 20 22:17 nginx
drwxr-xr-x 2 julio julio 4096 Jul 20 22:17 prometheus
```

Paso 6: Crear archivo de variables .env

1. Para subir al repositorio

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano .env.example
```

```
  GNU nano 7.2
MYSQL_ROOT_PASSWORD=nutrialianza2026
MYSQL_DATABASE=nutrialianza_db
MYSQL_USER=nutriapp
MYSQL_PASSWORD=nutriapp2026

GROQ_API_KEY=PEGAR_AQUI_TU_API_KEY_DE_GROQ
TELEGRAM_BOT_TOKEN=PEGAR_AQUI_TOKEN_DEL_BOT
TELEGRAM_CHAT_ID=PEGAR_AQUI_CHAT_ID

N8N_BASIC_AUTH_USER=admin
N8N_BASIC_AUTH_PASSWORD=admin2026
```

2. Variables reales con las APIs

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano .env
```

3. Cargar las variables

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ source .env
```

4. probar el token de telegram para ver si funciona

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitore
{"ok":true,"result":[]}julio@DESKTOP-JN2R04E:
```

Paso 7: Crear docker-compose.yml

es el archivo principal para levantar el stack del proyecto con Nginx, MySQL, Prometheus, Node Exporter, Loki, Filebeat y N8N.

1. Entramos al archivo Docker-compose.yml

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano docker-compose.yml
```

2. Validar que el archivo no tenga errores:

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose config
name: nutrialianza-monitoreo
services:
  filebeat:
    command:
```

3. Revisar que las variables del .env estén cargando

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ source .env
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ echo $MYSQL_DATABASE
nutrialianza_db
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ echo $N8N_BASIC_AUTH_USER
admin
```

Paso 8 — Configurar Nginx

1. Abrir el archivo de configuración de Nginx

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano nginx/default.conf
```

2.Crear la página web de prueba

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano nginx/html/index.html
```

Paso 9 — Crear base de datos inicial

1. Abrir el archivo de configuración de SQL

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano mysql/init/01_schema.sql
```

Paso 10 — Configurar Prometheus

Prometheus va a consultar métricas cada cierto tiempo. En el proyecto, Prometheus y Node Exporter se usan para capturar y almacenar métricas como CPU, RAM, red y disco. El flujo del proyecto indica que Prometheus almacena métricas y luego N8N puede consultarlas para generar alertas.

1. Abrir el archivo de configuración de Prometheus

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano prometheus/prometheus.yml
```

Paso 11 — Configurar Loki

1. Abrir el archivo de configuración de Loki

```
@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano loki/loki-config.yml
```

Paso 12 — Configurar Filebeat

1. Abrir el archivo de configuración de Filebeat

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ nano filebeat/filebeat.yml
```

- Filebeat queda leyendo los logs de Nginx y mostrándolos en consola. Luego, para la etapa intermedia, se puede mejorar la canalización completa hacia Loki.

Paso 13 — Levantar el stack

1. Validar que todo este creado

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ ls -R
.:
README.md  docker-compose.yml  filebeat  logs  loki  mysql  n8n  nginx  prometheus

./filebeat:
filebeat.yml

./logs:
nginx

./logs/nginx:

./loki:
loki-config.yml
```

2. También validemos el Docker-compose.yml

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose config
name: nutrialianza-monitoreo
services:
  filebeat:
    command:
```

3. Levantamos el stack

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose up -d
```

```
volumes:
[+] up 67/69:
[+] up 79/79fana/loki:2.9.8                                    Pulled
 ✔ Image grafana/loki:2.9.8                               Pulled
 ✔ Image nginx:latest                                     Pulled
 ✔ Image prom/node-exporter:latest                        Pulled
 ✔ Image docker.elastic.co/beats/filebeat:8.13.4          Pulled
 ✔ Image prom/prometheus:latest                           Pulled
 ✔ Image n8nio/n8n:latest                                 Pulled
 ✔ Image mysql:8.0                                        Pulled
 ✔ Network nutrialianza-monitoreo_nutrialianza-net        Created
 ✔ Volume nutrialianza-monitoreo_mysql_data               Created
 ✔ Volume nutrialianza-monitoreo_n8n_data                 Created
 ✔ Container nutrialianza-loki                            Started
 ✔ Container nutrialianza-mysql                           Started
 ✔ Container nutrialianza-node-exporter                   Started
 ✔ Container nutrialianza-prometheus                      Started
 ✔ Container nutrialianza-nginx                           Started
 ✔ Container nutrialianza-n8n                             Started
 ✔ Container nutrialianza-filebeat                        Started
```

Paso 14 — Verificar estado de los contenedores

1. Revisión de los contenedores

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose ps
     IMAGE                                    COMMAND                   SERVICE         CREATED            STATUS
     docker.elastic.co/beats/filebeat:8.13.4  "/usr/bin/tini -- /u…"   filebeat        4 minutes ago      Up 4 minutes
     grafana/loki:2.9.8                       "/usr/bin/loki -conf…"   loki            4 minutes ago      Up 4 minutes
     mysql:8.0                                "docker-entrypoint.s…"   mysql           4 minutes ago      Up 4 minutes
     n8nio/n8n:latest                         "tini -- /docker-ent…"   n8n             4 minutes ago      Up 4 minutes
     nginx:latest                             "/docker-entrypoint.…"   nginx           4 minutes ago      Up 4 minutes
```

Paso 15 — Probar Nginx

1. Devolver el HTML de la página de NutriAlianza

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ curl http://localhost
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <title>NutriAlianza S.A.</title>
</head>
<body>
    <h1>NutriAlianza S.A.</h1>
    <h2>Sistema de Monitoreo Inteligente con Docker e IA</h2>
    <p>Servidor web corporativo funcionando correctamente desde Docker + WSL2.</p>
    <p>Estado: OK</p>
</body>
</html>
```

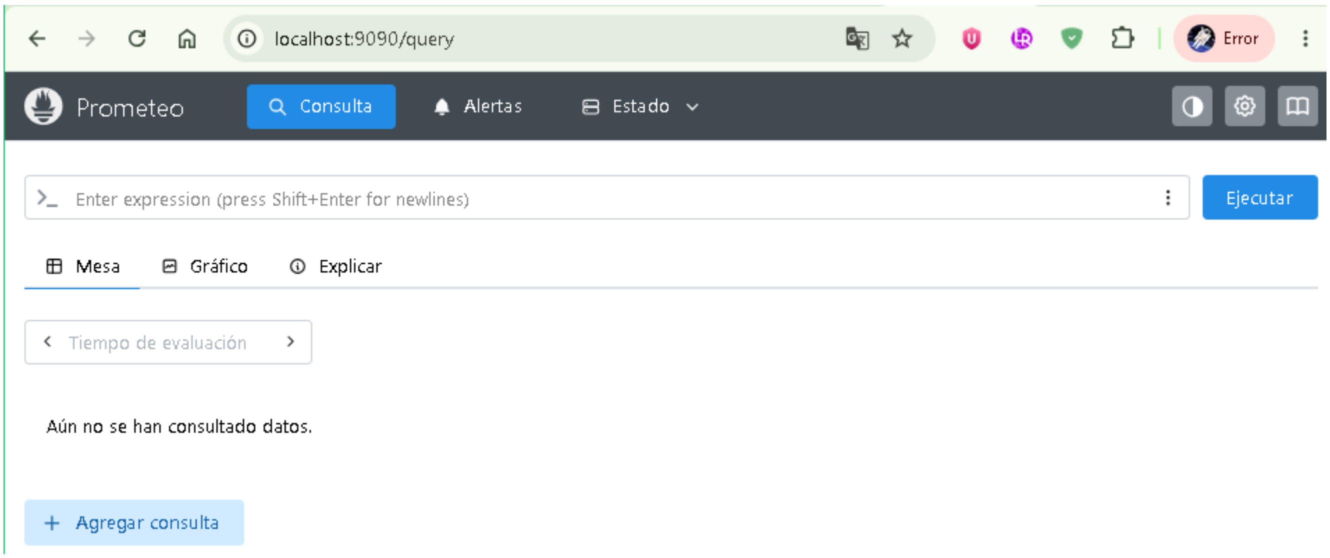
2. Mostrar el HTML en el navegador

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ curl http://localhost/health
NutriAlianza OK
```

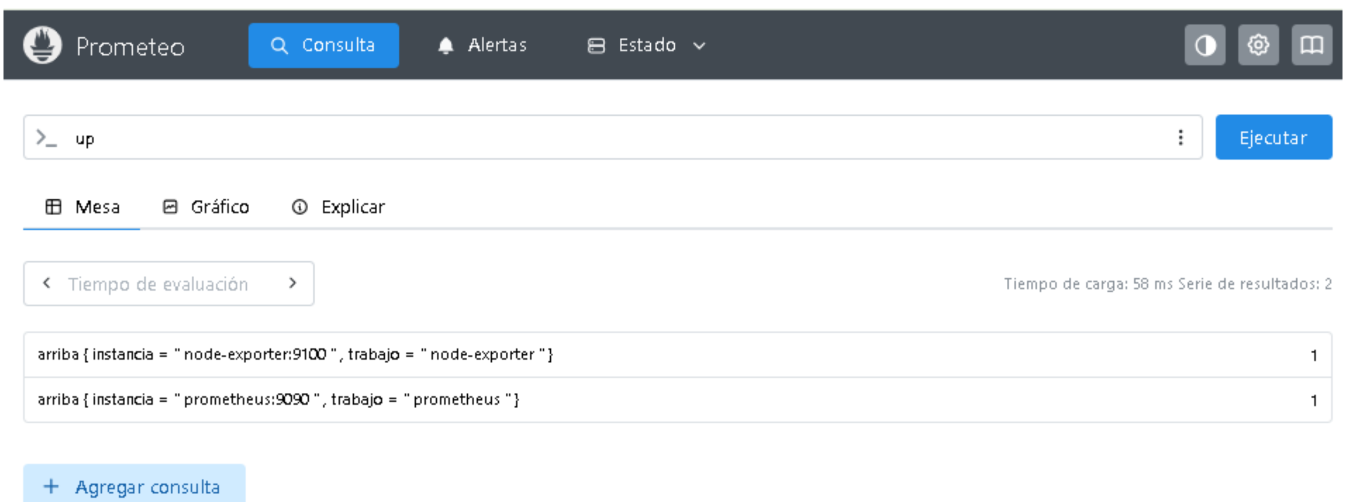
Paso 16 — Probar Prometheus

1. Probar servidor

2. Abrir en el navegador



3. Probar consulta y ejecutar



Paso 17 — Probar Loki

1. Revisando loki

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ curl http://localhost:3100/ready
ready
```

Paso 18 — Probar MySQL

1. Ver las bases de datos existentes

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose exec mysql mysql -u root -pnutrialianza2026 -e "SHOW DATABASES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| nutrialianza_db    |
| performance_schema |
| sys                |
+--------------------+
```

2. Verificamos que las tablas se hayan creado

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose exec mysql mysql -u root -pnutrialianza2026 -e "USE nutrialianza_db; SH
OW TABLES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+---------------------------+
| Tables_in_nutrialianza_db |
+---------------------------+
| formulas                  |
| inventario                |
+---------------------------+
```

3. Probamos que tenga datos la tabla de formulas

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose exec mysql mysql -u root -pnutrialianza2026 -e "USE nutrialianza_db; SE
LECT * FROM formulas;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+----+-------------------+----------------------+-------------+---------------------+
| id | nombre_formula    | ingrediente_principal | cantidad_kg | fecha_creacion      |
+----+-------------------+----------------------+-------------+---------------------+
|  1 | Formula Engorde A | Maiz                  |      500.00 | 2026-07-21 06:42:52 |
|  2 | Formula Lechera B | Soya                  |      350.00 | 2026-07-21 06:42:52 |
|  3 | Formula Avicola C | Trigo                 |      420.00 | 2026-07-21 06:42:52 |
+----+-------------------+----------------------+-------------+---------------------+
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$
```

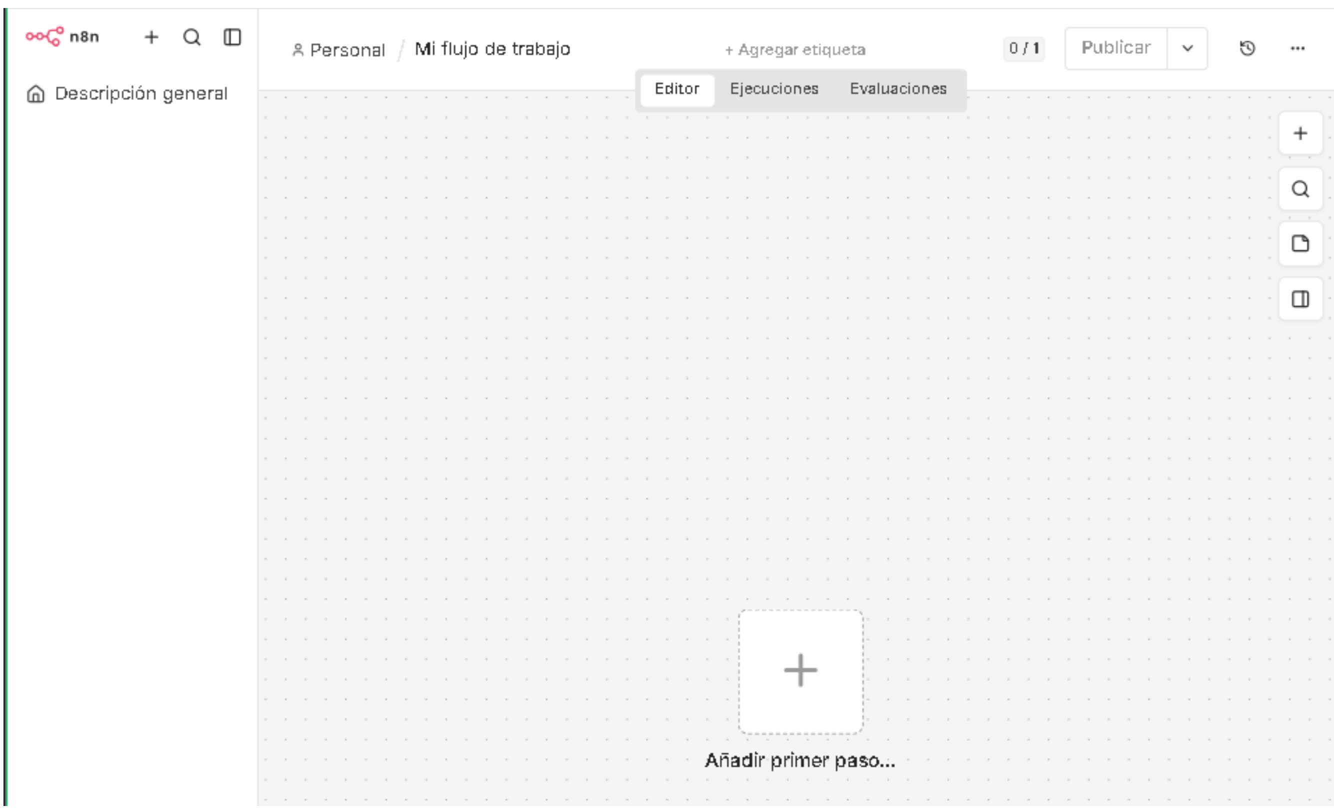
4. Probamos que tenga datos la tabla de inventario

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose exec mysql mysql -u root -pnutrialianza2026 -e "USE nutrialianza_db; SE
LECT * FROM inventario;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+----+----------+----------+----------------+
| id | producto | stock_kg | ubicacion      |
+----+----------+----------+----------------+
|  1 | Maiz     | 15000.00 | Bodega Central |
|  2 | Soya     |  9000.00 | Bodega Central |
|  3 | Trigo    | 12000.00 | Bodega Norte   |
+----+----------+----------+----------------+
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$
```

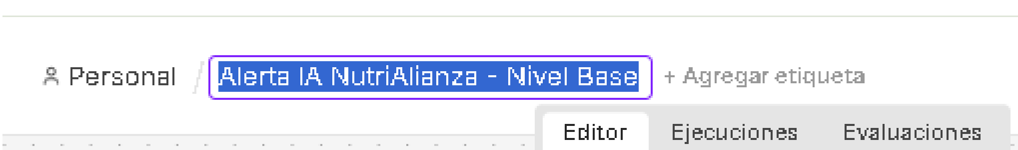
Paso 19 — Probar N8N

1. Abrí en navegador y creamos cuenta:

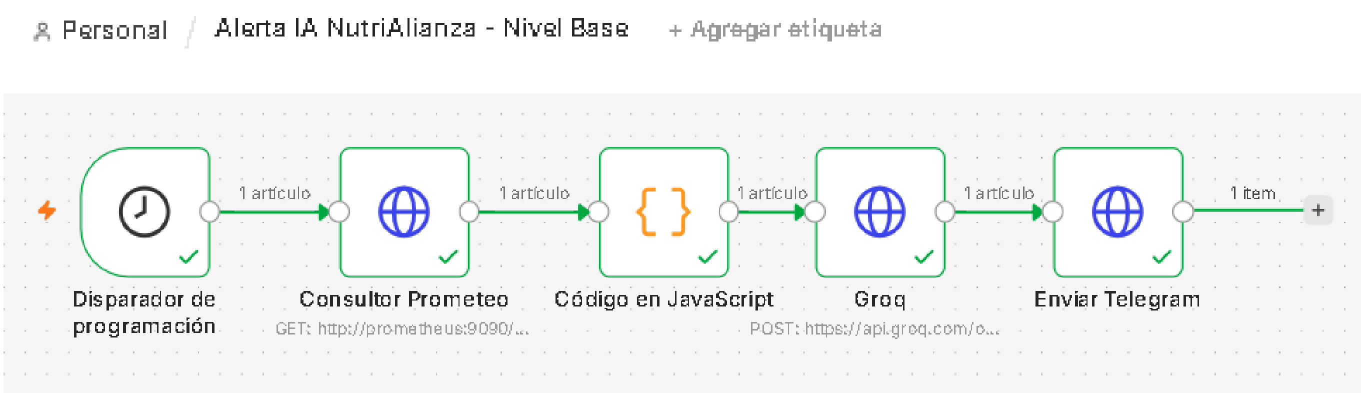
2. Damos clic en nuevo flujo de trabajo



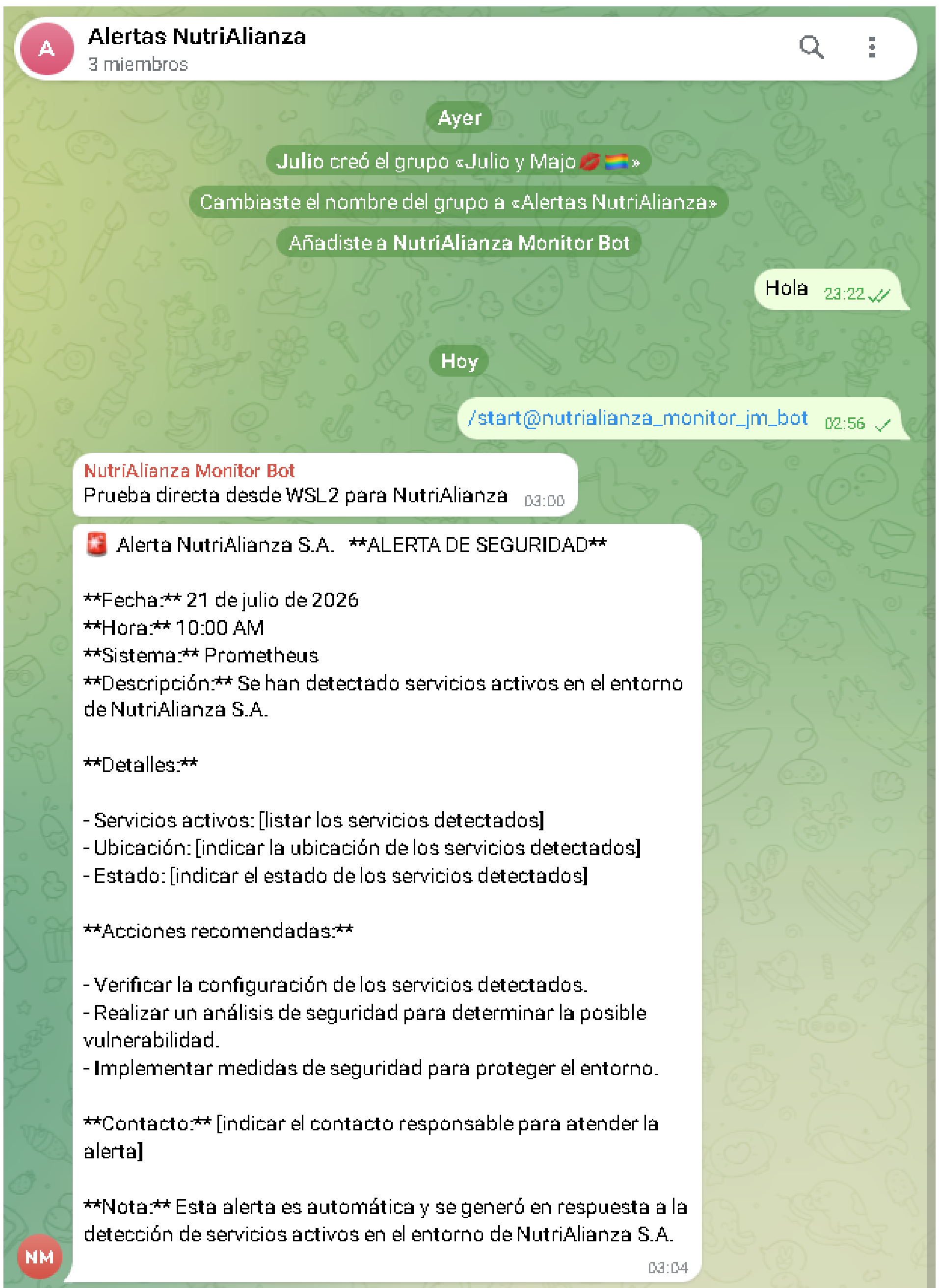
3. Cambiar el flujo del workflow



4. Agregar los nodos:



Paso 20 — Resultados de Telegram



Paso 21 — Revisión del (compose, UFW, Fail2ban)

1. Que todo esté levantado

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ docker compose ps
NAME                       IMAGE                                       COMMAND                  SERVICE          CREATED          STATU
S          PORTS
nutrialianza-filebeat      docker.elastic.co/beats/filebeat:8.13.4    "/usr/bin/tini -- /u…"   filebeat         2 hours ago      Up 2
hours
nutrialianza-loki          grafana/loki:2.9.8                          "/usr/bin/loki -conf…"   loki             2 hours ago      Up 2
hours      0.0.0.0:3100->3100/tcp, [::]:3100->3100/tcp
nutrialianza-mysql         mysql:8.0                                   "docker-entrypoint.s…"   mysql            2 hours ago      Up 2
hours      0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp
nutrialianza-n8n           n8nio/n8n:latest                            "tini -- /docker-ent…"   n8n              58 seconds ago   Up 43
 seconds   0.0.0.0:5678->5678/tcp, [::]:5678->5678/tcp
nutrialianza-nginx         nginx:latest                                "/docker-entrypoint.…"   nginx            2 hours ago      Up 2
hours      0.0.0.0:80->80/tcp, [::]:80->80/tcp
nutrialianza-node-exporter prom/node-exporter:latest                   "/bin/node_exporter"     node-exporter    2 hours ago      Up 2
hours      0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
nutrialianza-prometheus    prom/prometheus:latest                      "/bin/prometheus --c…"   prometheus       2 hours ago      Up 2
hours      0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$
```

2. UFW activo

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ sudo ufw status verbose
[sudo] password for julio:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
3306/tcp                   ALLOW IN    Anywhere
5678/tcp                   ALLOW IN    Anywhere
9090/tcp                   ALLOW IN    Anywhere
9100/tcp                   ALLOW IN    Anywhere
3100/tcp                   ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
3306/tcp (v6)              ALLOW IN    Anywhere (v6)
5678/tcp (v6)              ALLOW IN    Anywhere (v6)
9090/tcp (v6)              ALLOW IN    Anywhere (v6)
9100/tcp (v6)              ALLOW IN    Anywhere (v6)
3100/tcp (v6)              ALLOW IN    Anywhere (v6)
```

3. Fail2ban activo

```
julio@DESKTOP-JN2R04E:~/nutrialianza-monitoreo$ sudo systemctl status fail2ban --no-pager
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-07-20 21:23:59 CST; 5h 54min ago
       Docs: man:fail2ban(1)
   Main PID: 16778 (fail2ban-server)
      Tasks: 5 (limit: 4612)
     Memory: 20.8M (peak: 65.1M swap: 264.0K swap peak: 264.0K)
        CPU: 23.088s
     CGroup: /system.slice/fail2ban.service
             └─16778 /usr/bin/python3 /usr/bin/fail2ban-server -xf start
```